# Lecture 21: Pay Transparency

Compensation in Organizations

Jacob Kohlhepp

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# What You Are Paid vs. Who Knows What You Are Paid?

- Most of this class focuses on what people are paid.
- In this last lecture, we think about how much people know about how others are paid.
- Why might this matter?

# Discussion

How much will you be paid in your first job?

# The Pay of Economics Graduates

**BACHELOR'S DEGREE SALARIES BY MAJOR**

| ACADEMIC MAJOR | MEAN | 25TH PERCENTILE | MEDIAN | 75TH PERCENTILE | RESPONSES |
|---|---|---|---|---|---|
| Computer Engineering | $76,029 | $69,750 | $75,000 | $80,000 | 48 |
| Electrical Engineering | $75,558 | $69,500 | $72,000 | $77,500 | 55 |
| Engineering Technology | $72,933 | $68,000 | $72,000 | $79,000 | 17 |
| Environmental Engineering | $66,442 | $62,500 | $68,000 | $72,500 | 12 |
| Industrial/Manufacturing Engineering | $68,963 | $68,000 | $70,000 | $74,000 | 27 |
| Materials Engineering/Science | $71,382 | $69,500 | $72,000 | $75,000 | 23 |
| Mechanical Engineering | $72,825 | $68,500 | $71,000 | $75,000 | 53 |
| Nuclear Engineering | $72,000 | $69,500 | $71,500 | $75,000 | 3 |
| Petroleum Engineering | $85,260 | $69,500 | $80,000 | $105,000 | 5 |
| Software Engineering | $78,482 | $68,750 | $78,750 | $85,500 | 44 |
| Systems Engineering | $75,999 | $70,000 | $75,000 | $80,000 | 29 |
| HUMANITIES MAJORS | | | | | |
| Area & Gender Studies | $48,700 | $43,100 | $46,300 | $54,300 | 4 |
| English Language & Literature | $52,179 | $43,450 | $47,600 | $61,000 | 7 |
| Foreign Language & Literature | $48,700 | $43,100 | $46,300 | $54,300 | 4 |
| History | $49,332 | $45,000 | $47,600 | $51,860 | 5 |
| Liberal Arts/General Studies | $52,089 | $45,000 | $50,000 | $55,000 | 9 |
| Philosophy | $51,911 | $44,667 | $46,300 | $61,000 | 6 |
| Visual & Performing Arts | $48,700 | $43,100 | $46,300 | $54,300 | 4 |
| SOCIAL SCIENCES MAJORS | | | | | |
| Economics | $64,193 | $58,000 | $65,000 | $67,500 | 21 |
| Political Science/International Relations | $57,170 | $54,321 | $57,500 | $65,000 | 6 |

Source: NACE SALARY SURVEY (2022)

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Pay Transparency in the UNC System

Pay Transparency in General

# Discussion

The UNC System maintains a public salary database that is current to December 31, 2024.

# Pay Distribution by Job Class in Econ. Departments System-Wide

| JOB CATEGORY | INSTITUTION NAME | count | Min Salary | Med Salary | Max Salary |
|---|---|---|---|---|---|
| Professor (Primary) and Department Chair/Head | UNC-CH | 1 | 264313.0 | 264313.0 | 264313 |
| Professor (Primary) and Department Chair/Head | UNCG | 1 | 235289.0 | 235289.0 | 235289 |
| Department Chair / Head AC | NCSU | 2 | 221133.0 | 232522.5 | 243912 |
| Dean, Business | UNCP | 1 | 230308.0 | 230308.0 | 230308 |
| Associate Professor (Primary) and Department Chair/Head | UNCW | 1 | 229319.0 | 229319.0 | 229319 |
| Professor | UNC-CH | 17 | 120861.0 | 214479.0 | 469555 |
| Professor (Primary) and Department Chair/Head | ASU | 1 | 208276.0 | 208276.0 | 208276 |
| Professor | NCSU | 18 | 140319.0 | 197876.0 | 267056 |
| Professor | UNCG | 2 | 145346.0 | 181824.5 | 218303 |
| Assoc/Asst Dean, Business | UNCP | 1 | 176748.0 | 176748.0 | 176748 |
| Professor (Primary) and Department Chair/Head | ECU | 1 | 171668.0 | 171668.0 | 171668 |
| Associate Professor | UNC-CH | 8 | 104854.0 | 161234.5 | 186177 |
| Professor | UNCW | 6 | 94986.0 | 158320.0 | 183620 |
| Assistant Professor | UNC-CH | 11 | 97000.0 | 155324.0 | 164000 |
| Associate Professor | UNCC | 5 | 54596.5 | 154901.0 | 172291 |

The top 15 campus-job class pairs by median salary.

# Pay Distribution Econ Assistant Professors

| JOB CATEGORY | INSTITUTION NAME | count | Min Salary | Med Salary | Max Salary |
|---|---|---|---|---|---|
| Assistant Professor | UNC-CH | 11 | 97000 | 155324.0 | 164000 |
| Assistant Professor | UNCC | 5 | 84000 | 146520.0 | 149694 |
| Assistant Professor | WSSU | 1 | 136751 | 136751.0 | 136751 |
| Assistant Professor | NCSU | 14 | 88000 | 136369.0 | 149968 |
| Assistant Professor | UNCG | 3 | 110869 | 110869.0 | 110869 |
| Assistant Professor | NCA&T | 6 | 82000 | 110250.0 | 145400 |
| Assistant Professor | UNCW | 3 | 100000 | 107150.0 | 160760 |
| Assistant Professor | ECU | 2 | 98880 | 101840.5 | 104801 |
| Assistant Professor | UNCA | 2 | 95625 | 97448.5 | 99272 |
| Assistant Professor | ASU | 2 | 67000 | 88605.0 | 110210 |

# Salary Inversion Across Institutions

| JOB CATEGORY | INSTITUTION NAME | count | Min Salary | Med Salary | Max Salary |
|---|---|---|---|---|---|
| Associate Professor | UNC-CH | 8 | 104854.0 | 161234.5 | 186177 |
| Assistant Professor | UNC-CH | 11 | 97000.0 | 155324.0 | 164000 |
| Associate Professor | UNCC | 5 | 54596.5 | 154901.0 | 172291 |
| Associate Professor | WSSU | 1 | 151185.0 | 151185.0 | 151185 |
| Associate Professor | NCSU | 6 | 116875.0 | 147917.0 | 168244 |
| Assistant Professor | UNCC | 5 | 84000.0 | 146520.0 | 149694 |
| Associate Professor | UNCW | 7 | 126391.0 | 139053.0 | 169196 |
| Assistant Professor | WSSU | 1 | 136751.0 | 136751.0 | 136751 |
| Assistant Professor | NCSU | 14 | 88000.0 | 136369.0 | 149968 |
| Associate Professor | NCA&T | 5 | 118000.0 | 121800.0 | 130000 |

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Pay Transparency in General

Cullen (2024)

# Partial Equilibrium vs. General Equilibrium

- **Decision Problems:** Given everything is fixed, what happens if I do x vs. y?
  - Most everyday decisions.
  - Many people (including policymakers) consider.
  - Intuition/common sense works well here.

- **Games/Partial Equilibrium:** Given you respond to me, what happens if I do x vs. y if we fix most other things?
  - Most of this class (employer designs contract understanding worker will react)
  - Some people think this way for big decisions (college, job offers, etc.)
  - Some government policy consider this (Ledbetter case)
  - Intuition/common sense sometimes works well here.

- **General Equilibrium:** Given you respond to me and everything around us also responds, what happens if I do x vs. y?
  - This class largely ignored this.
  - People almost never think this way (and this is largely rational).
  - With the exception of macroeconomists, many economists abstract from this.
  - Intuition/common sense are often wrong here.
  - But this matters for pay transparency, because we change the info. everyone has.

# The Ledbetter Case and “Decision Problem” Thinking

- Ledbetter learns the salary of a coworker doing similar work.
- It turns out this was true of the other (male) coworkers.
- Ledbetter presses charges against her company.
- Decision problem thinking works here, because this is a single company and one employee.
- This is also “horizontal” pay transparency.

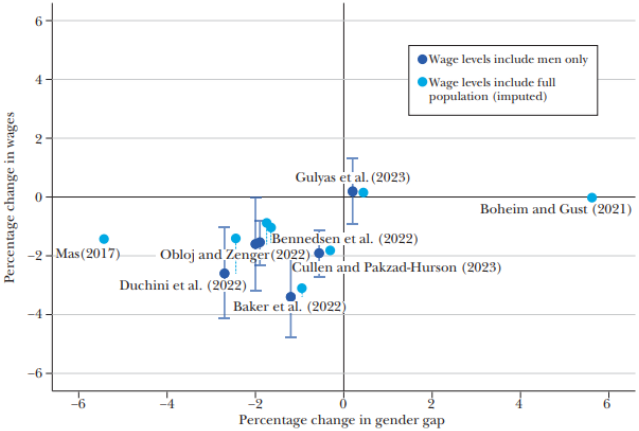
# The 2009 Lilly Ledbetter Fair Pay Act

- People have tried to use Ledbetter as an example for larger reforms.
- But scaling the logic from one case can be dangerous.
- There can be partial and general equilibrium effects.
- Question: What does this mean?

# Horizontal Pay Transparency Has Unintended Consequences

*Figure 1*

**Effect of Pay Transparency on Wages, Existing Studies**




*Source:* Each observation is a study of a pay transparency policy, with study details and results describe further in Table 2.
*Note:* The y-axis is the percentage change in wages, and the x-axis represents the percentage change in the gender wage gap. Dark blue dots represent the percentage change in wages for men only, with standard error bars, while light blue dots represent the percentage change in wages for all workers.

# Unintended Consequence 1: A Partial Equilibrium Effect

- One explanation relates directly to this class.
- When a worker sees they are paid less than peers, they become discouraged.
- They then exert less effort and are less productive.
- The firm then pays even lower wages.
- In what ways it this similar to the logic of relational contracts?
- How is it not?

# Unintended Consequence 2: A General Equilibrium Effect

- Consider an organization with $100$ workers with wage $x$ before pay transparency.
- If one of the workers asks for a raise of $x + 1$, the firm can grant the raise at cost of $1$.
- After pay transparency, if the worker asks for a raise of $1$ the cost is now $100$!
- The firm will now negotiate more aggressively against requests for raises.
- And importantly, the firm can credibly say "If I give it to you, I must give it to everyone."
- Finally, workers will not negotiate aggressively because they could be fired, and the benefit of a raise is shared.

# Discussion

What is the first-year total compensation of an economics PhD at a consulting firm?

# Vertical Pay Transparency (Cullen and Perez-Truglia (2022))

- Employees underestimated the average salary of their manager's position by 14%
- This inflates future expectations about their own earnings.
- Employees responded by increasing sales by 1.1%, emails by 1.3%, and hours by 1.5% for each 10% increase in the perceived manager's pay.
- This is a productivity improving, career-concerns-like-effect.